Cross-Reader Generality of PAQ-KV: A Multi-VLM Study (Internal Companion Note)

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Abstract

PAQ-KV is a training-free, index-free, single-reader method: a vision–language model (VLM) prefills a multi-page document once, and per query computes its *own* question-to-context attention, selects the top $\approx$5% of 64-token visual key/value (KV) blocks, and decodes under a reversible 4-D restricted-attention mask over its own KV cache. Its headline result—beating a trained ColQwen2 retriever on MMLongBench-Doc accuracy—is established on Qwen3-VL-8B and confirmed out-of-sample. This note asks the natural follow-up: *does the advantage generalize to other VLM readers?* We port the full method (reader-own scoring, selection, and masked decode) to additional VLMs and report matched-budget results on a fixed 80-document / 444-query development (DEV) pin. Findings so far: on Qwen3-VL-4B (same architecture, half the parameters) the method largely *holds* at byte-identical budget ($\Delta = +0.030$, $\mathrm{LB}_{95} = -0.004$, a near-win that denoises above full context); on Qwen2.5-VL-7B (same Qwen-VL family) it is a *statistical dead heat* ($\Delta = +0.007$, $\mathrm{LB}_{95} = -0.037$); on InternVL3.5-8B (a different VLM whose language backbone is Qwen3) it *significantly underperforms* ($\Delta = -0.067$, $\mathrm{LB}_{95} = -0.106$). Together these characterize a **mechanistic boundary** located at architecture rather than scale: the method's benefit tracks how good the host reader's own attention is as a below-page retriever. On a second DEV benchmark (LongDocURL) all three readers are at or below their MMLongBench-Doc result—Qwen3-VL-4B ties, the other two are significant negatives—preserving the ordering on the harder, conversion-bound cell (§2.3). Three further readers (Molmo-7B-D, MiniCPM-V-4.5, PaliGemma-2) were each found incompatible with the current setup for distinct, honest reasons (remote-code version drift for the first two; an $\approx$8K context that cannot hold a multi-page document for the third), so the study's evidence rests on the four readers above. This is an internal companion note kept separate from the submission; the shared team decides what, if anything, to fold in.

1 Setup

1.1 The method and the established result

PAQ-KV prefills a full multi-page document once into a VLM's KV cache. For each query it (i) reads the reader's own question-to-context attention (scorer v0: mean over heads and layers), (ii) partitions the retained visual KV into 64-token stripe blocks and selects the top $\approx$5% by attention score, and (iii) decodes the answer under a reversible 4-D additive attention mask that hides all non-selected visual columns while always keeping the (small) non-visual scaffold. It trains nothing, builds no external index, and uses a single model.

On **Qwen3-VL-8B**, out-of-sample on a sealed 45-document / 244-query MMLongBench-Doc split, PAQ-KV at 5% scores **0.3957** versus a trained ColQwen2 retriever's **0.3202** ($+0.0755$; doc-cluster $\mathrm{LB}_{95} = +0.0290$). On the 80-document / 444-query DEV pin the same beat is $+0.079$ ($\mathrm{LB}_{95} = +0.041$). This note does not touch that result; the sealed split remains evaluated exactly once.

1.2 What generality requires of a reader

Porting surfaced hard prerequisites that any candidate reader must satisfy for the method to even run:

1. **White-box access.** The method reads attention weights, edits the KV cache, and injects a 4-D mask. Any API / closed-weights model is categorically impossible.
2. **In-line visual tokens in the decoder's self-attention KV.** Blocks are partitions of visual token *positions in the self-attention sequence*. A reader that injects vision via *cross-attention* (vision lives outside the self-attention KV) has no visual KV blocks to select or mask.
3. **Hookable eager attention returning 4-D weights**, plus a **4-D additive mask honored by the language model** (the transferability crux; native TRANSFORMERS classes that delegate to a native decoder satisfy this, custom trust_remote_code attentions may not).
4. **Reconstructable, continuable positions** (1-D arange or M-RoPE) and enough context to prefill a multi-page document. Sub-native context ceilings force a reduced-budget setting (a disclosed confound, §4).
5. Empirically (§3), **the reader's own attention must actually be a good below-page retriever.** Mechanical portability is necessary but not sufficient.

1.3 Protocol

All second-reader numbers are on the fixed **80-document / 444-query DEV pin** (paq_confirm_pin.json, MMLongBench-Doc DEV), reported as a *generality control*, not a new sealed claim. Each reader is compared to itself under two selectors at a matched budget: PAQ-KV (own attention selects $\approx$5% of blocks) vs. a *comparator* arm in which the same reader reads the top-5% pages chosen by cached ColQwen2 retrieval scores (no new ColQwen2 runs). We also record a *full-context* reference. The primary statistic is the doc-cluster 95% lower bound LB_{95} of (paqkv − comparator) (seed 0, 10,000 bootstrap resamples); the pre-registered "transfers" bar is $\mathrm{LB}_{95} > 0$. An adversarial worst-case imputation of any missing query binds the verdict. A hard **identity gate** (teacher-forced argmax parity of the all-true 4-D mask vs. native causal decode, plus a restrictive mask changing the output, on $\geq$2 distinct documents including one $\geq$30 pages) must pass before any number is trusted. Verdicts are labeled INCOMPATIBLE (identity fails / port infeasible), INCOMPLETE (coverage/power short), READER2_NO_TRANSFER ($\mathrm{LB}_{95} \leq 0$ with clean coverage), or READER2_TRANSFERS ($\mathrm{LB}_{95} > 0$).

2 Results

2.1 Main comparison (verified readers)

Table 1 reports the three verified readers on the same 444-query DEV pin. Accuracy is the MMLongBench-Doc scorer; higher is better. All three readers read the same content selectors (own PAQ-KV blocks vs. ColQwen2 pages) at a matched $\approx$5% budget.

- **Qwen3-VL-8B (native reader): win.** PAQ-KV − ColQwen2 $= +0.076$, $\mathrm{LB}_{95} = +0.041$; PAQ-KV even edges its own full context (0.4042 vs. 0.3965), i.e. selection *denoises*.
- **Qwen3-VL-4B (same architecture, half the parameters): near-win / method holds.** $\Delta = +0.030$, $\mathrm{LB}_{95} = -0.004$ (just misses the bar at $n = 444$; the point estimate is a clear positive). PAQ-KV again *denoises* above its own full context (0.3640 vs. 0.3480), and on the 137 answerable queries it recovers **0.869** vs. the comparator's **0.720**—the strongest below-page selection of any second reader. Per-query: 63 wins / 336 ties / 45 losses. Because the input geometry is byte-identical to the 8B (verified on all 80 documents), the gap to the 8B (-0.040 cross-reader) is a pure capacity effect, not a budget artifact. The method's mechanism clearly survives shrinking the reader within the same architecture.
- **Qwen2.5-VL-7B (same Qwen-VL family): tie.** $\Delta = +0.0074$, $\mathrm{LB}_{95} = [-0.037, +0.049]$—a statistical dead heat straddling zero (PAQ-KV nominally ahead). Verdict READER2_NO_TRANSFER by the $\mathrm{LB}_{95} > 0$ bar, but this is "no detectable difference," not a loss. Per-query: 54 wins / 331 ties / 59 losses. On the 128 queries answerable by full context, PAQ-KV recovers **0.703** vs. the comparator's **0.581**—here PAQ-KV is the *better* below-page selector. It loses only -0.054 $[-0.087, -0.024]$ to its own full context.
- **InternVL3.5-8B (cross-family, Qwen3 LM): significant negative.** PAQ-KV − ColQwen2 $= -0.067$, $\mathrm{LB}_{95} = [-0.106, -0.029]$ (whole interval below zero). It also trails its own full context (-0.076 $[-0.113, -0.041]$). On answerable queries it recovers only **0.529**, losing $\approx$47% of them to its 8% selection. Verdict READER2_NO_TRANSFER (a genuine, well-powered negative).

2.2 Additional readers (in progress)

Four further readers were requested for this study. Qwen3-VL-4B is **complete** and appears in Tables 1 and 4 (it is the same TRANSFORMERS class as the sealed 8B, so it reused the audited harness with a loader swap and reproduced the 8B's byte-identical input geometry). The remaining three (Table 2) each require a genuine per-model port because their attention APIs differ from the standard TRANSFORMERS 4-D-mask path the method relies on. Feasibility is decided cheaply by a Phase-0 identity spike before any full run; full ports proceed only for spike-passers, and a reader that fails the identity gate is reported honestly as INCOMPATIBLE rather than forced.

Per-reader notes.

- **Qwen3-VL-4B** is the same architecture as the sealed 8B (same TRANSFORMERS class, M-RoPE, native per-image budget), so the existing harness runs with a loader swap. It is a *within-architecture size ablation*, not a cross-architecture test; a win here would say the method survives shrinking the same reader, not that it transfers across families.
- **Molmo-7B-D** $\rightarrow$ INCOMPATIBLE (version drift, not mechanism). Its custom attention *does* expose the exact injection point the method needs—the top-level forward accepts a 4-D attention_bias fed straight to SDPA—so the mechanism is portable in principle. But its 2024 remote-code modeling file will not load under the pinned TRANSFORMERS 5.5.4: a cascade of tied-weights API mismatches (all_tied_weights_keys, then tie_weights(missing_keys=...)) blocks from_pretrained. Loading it would require porting Molmo's modeling file to the current API, out of scope here. This is an engineering/version-compatibility incompatibility, not a statement about the method.
- **MiniCPM-V-4.5** $\rightarrow$ INCOMPATIBLE (version drift, not mechanism). The model weights load, but its 2025 remote-code *processor* cannot build inputs under the pinned TRANSFORMERS 5.5.4: it expects a custom tokenizer attribute (tokenizer.im_start_id) that the current fast-tokenizer backend does not expose. Independently, its 3D-Resampler compresses each page to $\approx$64 tokens, so even a working port would be near-degenerate below page granularity (a 5% block budget of $\approx$64 tokens is < 1 block per page); and its LM is Qwen3, the same family as the InternVL negative.
- **PaliGemma-2** $\rightarrow$ INCOMPATIBLE (representation, not version). It loads natively and accepts multiple images (1024 tokens/page), but its Gemma-2 language model has an $\approx$8192-token context with a 4096 sliding window, and PaliGemma uses a *fixed* 1024 tokens per image with no dynamic per-page budget. That caps a document at $\approx$7

Table 1: PAQ-KV vs. the ColQwen2-page comparator, within each reader, on the 80-doc / 444-q DEV pin. Δ = mean(paqkv − ColQwen2); LB_{95} = doc-cluster 95% lower bound (pre-registered "transfers" bar is $\mathrm{LB}_{95} > 0$). "ctx" is the mean realized prefill length. Qwen3-VL-8B is the reader the method was developed and sealed on; Qwen3-VL-4B uses byte-identical input geometry (a pure size ablation).

Reader	ctx	PAQ-KV	ColQwen2	Full	Δ	LB_{95}
Qwen3-VL-8B *(native, sealed)*	∼69K	**0.4042**	0.3284	0.3965	+0.076	+0.041
Qwen3-VL-4B *(size ablation)*	∼69K	0.3640	0.3336	0.3480	+0.030	−0.004
Qwen2.5-VL-7B	∼52K	0.2698	0.2623	0.3237	+0.007	−0.037
InternVL3.5-8B	∼36K	0.1876	0.2542	0.2637	−0.067	−0.106

Table 2: Additional candidate readers. "Native" = a TRANSFORMERS 5.5 class exists (mask machinery and eager hooks transfer as for InternVL/Qwen2.5-VL); "remote" = trust_remote_code custom attention (4-D mask honored and hookability must be validated by the spike). Results are filled in as runs complete.

Reader	LM backbone	Vision integration	Impl.	Phase-0 verdict
Molmo-7B-D	Qwen2-7B	CLIP, in-line tokens	remote	INCOMPATIBLE (model won't load: version drift)
MiniCPM-V-4.5	Qwen3-8B	3D-Resampler (64 tok)	remote	INCOMPATIBLE (processor won't build inputs)
PaliGemma-2	Gemma-2	prefix-LM (bidir.)	native	INCOMPATIBLE (∼8K ctx: cannot fit a doc)

pages; the study's documents are 17–104 pages (median ≈32), so PAQ-KV's "prefill the whole document once" premise cannot be satisfied—the document does not fit. (Its prefix-LM attention and short-output tuning would further complicate a port even at fitting lengths.)

Takeaway from the three incompatibilities. They are not method failures; they map the method's *practical* deployment envelope. Running reader-own PAQ-KV needs, in addition to the mechanistic property of §3: (i) a version-current implementation exposing a hookable attention and an honored additive mask; and (ii) enough context, with a dynamic per-page token budget, to prefill a whole multi-page document. Among the readers surveyed, the Qwen-VL family (Qwen3-VL-8B/4B, Qwen2.5-VL-7B) and InternVL3.5-8B satisfy both; the three above each miss one. This is worth stating plainly as a scope condition rather than omitting it.

2.3 Second benchmark: LongDocURL-DEV

For completeness we ran all three readers on the second DEV benchmark, LongDocURL (80 docs / 463 q; longer documents, median 44 vs. 28 pages), under the identical protocol (reader-own PAQ-KV vs. a matched ColQwen2-page comparator, same doc-cluster LB_{95}, same identity gate). One harness note: InternVL's identity gate was aligned to the teacher-forced mask_faithful criterion the other readers already use—on a long document a benign near-tie in the multi-chunk prefill had tripped the older greedy string check, while teacher-forced argmax confirms the mask is faithful (0 argmax changes, max $|\Delta\text{logit}| = 0$). Table 3 pairs the two benchmarks.

Every reader is READER2_NO_TRANSFER on LongDocURL, and every reader is at or below its MMLongBench-Doc result on this harder cell. Qwen3-VL-4B narrows to a tie (Δ+0.006, LB_{95}−0.028); Qwen2.5-VL-7B and InternVL3.5-8B are significant negatives (−0.078 and −0.090; both intervals below zero). The reader ordering is preserved (4B > Qwen2.5-VL > InternVL) and the whole gradient shifts down. This is consistent with LongDocURL being reader-conversion-bound—even the native Qwen3-VL-8B method only reaches *parity* there (no significant beat)—and with the mechanistic reading (§3) that reader-own selection helps least where the answer is harder to convert from the retained evidence.

3 Discussion: a mechanistic generality gradient

The verified readers order cleanly by architectural distance from the reader the method was developed on (Table 4). The benefit is not universal; it *degrades monotonically* as the reader moves away from Qwen3-VL-8B, and the degradation tracks a measurable property—how well the reader's own question-to-context attention localizes the answer below page granularity. Qwen3-VL-8B has this property (it recovers answerable queries well at 5% and even denoises above full context); **Qwen3-VL-4B retains it** (near-win at matched, byte-identical budget; denoises above full; recovers 0.87 of answerable queries), so the effect survives halving the parameter count *within the same architecture* and is not an artifact of the 8B's scale alone; Qwen2.5-VL-7B partially has it (break-even; recovers 0.70); InternVL3.5-8B lacks it here (recovers 0.53 and loses). The two Qwen3-VL points (win, near-win) versus the same-family tie and the cross-family negative locate the boundary at *architecture*, not merely scale. This *confirms* rather than refutes the single-reader observation, and reframes it from an unexamined weakness into an investigated boundary condition: PAQ-KV is a white-box method whose benefit is contingent on the host reader's attention being an effective below-page retriever.

Table 3: Both DEV benchmarks, all three readers (444-q MMLongBench-Doc; 463-q LongDocURL; full coverage, identity pass). Δ / LB_{95} are doc-cluster (paqkv − comparator). Every reader is at or below its MMLongBench-Doc result on the harder LongDocURL cell; the reader ordering is preserved.

	MMLongBench-Doc (DEV)			LongDocURL (DEV)		
Reader	PAQ-KV	comp.	Δ / LB_{95}	PAQ-KV	comp.	Δ / LB_{95}
Qwen3-VL-4B	0.3640	0.3336	+0.030 / −0.004	0.4966	0.4908	+0.006 / −0.028
Qwen2.5-VL-7B	0.2698	0.2623	+0.007 / −0.037	0.3953	0.4728	−0.078 / −0.116
InternVL3.5-8B	0.1876	0.2542	−0.067 / −0.106	0.2186	0.3081	−0.090 / −0.127

Table 4: The generality gradient, ordered by architectural distance from the sealed reader. Δ / doc-cluster LB_{95} (paqkv − comparator), DEV pin (Qwen3-VL-8B also shows a sealed out-of-sample $LB_{95} = +0.029$). The benefit degrades monotonically with distance.

Reader	Relation to the sealed reader	Δ / LB_{95}
Qwen3-VL-8B	native (sealed reader)	+0.076 / +0.041 *(win)*
Qwen3-VL-4B	same architecture, 4B	+0.030 / −0.004 *(near-win)*
Qwen2.5-VL-7B	same family, prior gen.	+0.007 / −0.037 *(tie)*
InternVL3.5-8B	cross-family VLM (Qwen3 LM)	−0.067 / −0.106 *(neg.)*

3.1 Can parameter tuning close the InternVL gap? (DEV-only, pre-registered, held-out)

A natural follow-up: is InternVL's deficit an artifact of the fixed 5% budget or the default attention read-out, and can tuning close it? We answered this under a frozen pre-registration that never touches the sealed set: a TUNE(40)/HELDOUT(40) document split of the DEV pin (seed 0, stratified, positive-allowlist enforced), a frozen grid over budget $R \in \{5, 10, 20, 30\}\%$, attention read-out {mean-all, last-quarter-layers, max-over-heads, entropy-weighted}, token→block aggregation, and a boundary-keep policy (measured in *attended visual tokens*), with the winning configuration selected on TUNE and evaluated once on HELDOUT. A pre-GPU power analysis fixed the reading: at n=40 documents the doc-cluster half-width is ∼0.06, so "matching" cannot register as superiority—hence the primary is the descriptive budget frontier (all 80 docs) plus the low-variance paired C^*−default contrast and a noninferiority margin δ=0.03.

Budget is the only lever, and it does not yield a win. The frontier (Table 5) shows InternVL's gap to its comparator narrowing monotonically with budget (−0.067 at 5% to −0.001 at 30%), and the method reaching its *own full context* by ∼20% (recovery of answerable queries rises from 0.53 at 5% to 0.74 at 10%). But LB_{95}(method − comparator) stays below zero at *every* budget—it never becomes a significant win. The tuned winner C^* (max-over-heads read-out, R^*=10%) gives **no gain beyond budget** on HELDOUT (paired C^*−default: Δ+0.016, LB_{95} −0.014), and is **not noninferior** to the comparator at R^* (Δ−0.006, LB_{95} $-0.055 < -\delta$), despite the method using *more* attended visual tokens than the comparator there (2126 vs. 1768). So a smarter attention read-out does not rescue InternVL: only spending several times the budget narrows the within-reader gap, and even then only to break-even with its own full context, never to a comparator win. This *reinforces* §3: InternVL's own attention is a genuinely weaker below-page retriever, and this cannot be tuned away with these knobs. (All values are DEV held-out estimates, pre-registered; the sealed set is untouched and no sealed claim is made.)

4 Confounds and honesty caveats

Any use of these numbers must carry the following.

1. **Reduced-budget confound.** Both second readers run below the native-long 8B budget (InternVL ∼36K via a 40K context ceiling; Qwen2.5-VL ∼52K via a 128K ceiling plus a processor pre-resize). Within-reader matched-budget comparisons carry the findings; cross-reader comparisons *against* the 8B are confounded by capacity and budget and are secondary only.
2. **Shared LM family.** InternVL3.5-8B's language backbone is Qwen3, which is *why* the KV/attention/mask machinery transfers. It is a legitimate cross-VLM result but not a fully independent language architecture; disclose it.
3. **Conservative-against-method budget asymmetry (Qwen2.5-VL).** The method downscales all pages in the full-document prefill, whereas the comparator renders its 1–5 kept pages near-native and thus gets a slightly larger per-page budget. The tie is therefore a floor on the within-family effect.
4. **DEV-pin, not a new sealed claim.** All second-reader numbers are on the DEV pin as a generality control. The sealed split is untouched.

Verification (Appendix A)

Every reported run passed a hard identity gate and was independently checked. For each verified reader: the identity audit confirmed the all-true 4-D mask reproduces native causal decoding under the teacher-forced argmax criterion (max $|\Delta\text{logit}| = 0$), a restrictive mask changes

Table 5: InternVL3.5-8B budget frontier on all 80 DEV documents (default read-out; descriptive, no selection). Δ, LB_{95} are doc-cluster (method − comparator). The gap narrows with budget but LB_{95} never clears zero; the method reaches its own full context by $\sim$20%.

R	method	comparator	full	Δ(m−c)	LB_{95}(m−c)
5%	0.1876	0.2542	0.2637	−0.067	−0.106
10%	0.2367	0.2762	0.2637	−0.040	−0.083
20%	0.2664	0.2767	0.2637	−0.010	−0.048
30%	0.2684	0.2691	0.2637	−0.001	−0.036

the output, on $\geq$2 distinct documents (one $\geq$30 pages), with the resolved model revision and code hash bound into the frozen run manifest. A CPU sparse-mask audit confirmed the block-to-column mapping is exact (all-blocks selection reproduces the all-true mask; a sparse subset masks exactly the intended columns). An off-meter code reviewer independently attempted to falsify each result and found no implementation defect (“genuine negative” for InternVL; “genuine null with favorable caveats” for Qwen2.5-VL). Full coverage for both second readers: 444/444 queries, 80 documents, 0 skipped, 0 dropped, identity pass. Artifacts, verdict JSONs, reviewer transcripts, and audit scripts are in the project repository under research/ and research/reviews/paq_second_vlm_2026-07-22/; the consolidated running log is research/design_notes/paq_cross_reader_generality_2026-07-23.md.

References (model / benchmark sources)

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